David Moore

Concord, NC • 508-769-9925 • hello@david-moore.me

Staff Software Engineer with 20+ years of Python 3 backend and AWS cloud platform experience, specializing in security-by-design systems, SOC 2 compliance programs, developer experience tooling, and operational excellence. Currently driving platform security automation and infrastructure-as-code at Optimal Dynamics, following nearly 7 years as a proven hands on technical leader at NewtonX, with a track record of building reliable, observable, and maintainable infrastructure while mentoring engineers and driving platform strategy.

SUMMARY OF ACCOMPLISHMENTS AND CORE STRENGTHS

- **Security & Compliance:** Owned NewtonX's entire cybersecurity program — maintained SOC 2 compliance through 3 consecutive annual audits and ensured ongoing GDPR adherence. Built the company's zero trust solution (Cloudflare + Sophos) from scratch, reducing non-compliant devices from hundreds to under 40. Migrated all systems to SSO & OIDC, enforced security training, and championed security-by-design across the engineering org.
- **Python 3 & AWS Platform Engineering:** Deep Python 3 expertise (Django, Celery patterns) across high-throughput production systems. AWS experience spanning compute, storage, queuing, and database tiers. Proficient with Terraform for infrastructure-as-code, Kubernetes and Helm for container orchestration, and CI/CD pipelines (Jenkins, Cloud Build, CodePipeline-equivalent patterns).
- **Operational Excellence & Observability:** Designed the engineering team's gold-standard on-call and incident response process from scratch. Reduced spurious auto-open incidents by 90% through a systematic operational excellence initiative. Championed monitoring, alerting, and observability as first-class requirements for every platform service shipped.
- **Developer Experience & Tooling:** Improved CI/CD pipelines by generating Helm charts for Kubernetes deployments, enforcing unit testing standards in pull requests, and helping migrate hand-built cloud projects to Terraform. Built internal tooling that accelerated developer workflows. Led migration from self-hosted infrastructure (Jenkins, RabbitMQ, Elasticsearch) to managed SaaS equivalents to reduce toil.
- **Event-Driven Architecture & Distributed Systems:** Designed and delivered high-throughput event-driven platforms using Kafka, RabbitMQ, and Celery — scaling async processing pipelines from 200K to 1M+ operations/day. Architected distributed worker systems for fault tolerance, horizontal scalability, and operational simplicity.
- **AI Tooling & Technical Leadership:** Led team-wide adoption of AI-assisted development tools (Anthropic Claude, GitHub Copilot) to accelerate refactoring, test generation, and prototyping. Mentored engineers at all levels; one grew to lead a sibling team. Participated in recruiting, hiring, and onboarding. Drove cross-team technical initiatives and established engineering best practices.

TECHNICAL SKILLS

Python 3 • Django • Celery • Go • Java (Spring Boot) • AWS (ECS/SQS/RDS/S3/DynamoDB) • GCP • Terraform • Spacelift • Bazel • Kubernetes • Helm • Docker
Kafka • RabbitMQ • Pub/Sub • Event-Driven Architecture • Distributed Systems • Microservices • REST APIs
Trino • Dagster • dbt • DuckDB • Preset • React • Redux • TypeScript
SOC 2 • GDPR • Zero Trust • SSO • OIDC • Auth0 • Vanta • Security Hardening • Cloudflare • Compliance Frameworks
PostgreSQL • DynamoDB • Firestore • Redis • Elasticsearch • BigTable • GitHub • Jenkins • CI/CD • Observability
IC Technical Leadership • Mentoring • Agile/Scrum • Incident Response • AI-Assisted Development (Claude, Copilot)

PROFESSIONAL EXPERIENCE

Optimal Dynamics, New York, NY; *Staff Software Engineer, Backend* April 2026–Present
Joined a Series C logistics/supply-chain optimization startup to strengthen platform security automation, observability, and infrastructure-as-code practices.

- Building an agentic AI-based workflow to triage and remediate dependencies flagged for known security vulnerabilities, reducing manual patch turnaround.
- Leading evaluation of observability platforms (Honeycomb, Dash0, Grafana, self-hosted SigNoz, etc.) to replace ad hoc tooling and build solid observability for the organization.
- Untangling uneven Spacelift adoption by migrating more Terraform modules into managed Spacelift stacks and resolving a year of accumulated drift across disaster-recovery infrastructure.

NewtonX, Inc., New York, NY; *Senior Staff Software Engineer / Senior Engineering Manager* October 2019–April 2026
Took full technical ownership of platform infrastructure, security, and developer experience at a high-growth B2B SaaS startup. Designed and operated Python-based cloud-native systems on GCP and AWS, ran the company's compliance program, and established the engineering practices that powered the platform's reliability and growth.

Security, Compliance & Zero Trust

- Owned the company's end-to-end cybersecurity program: maintained SOC 2 compliance through 3 consecutive annual audits, ensured GDPR adherence, and enforced security training and background check policies across the organization.
- Built the company's zero trust solution from scratch (Cloudflare + Sophos), reducing non-compliant devices from hundreds to under 40. Designed and implemented the device compliance verification system and a scalable remediation workflow for out-of-compliance employees and contractors.
- Migrated all internal systems from per-service authentication to centralized SSO and OIDC, ensuring consistent password and session policies across the org. Integrated Auth0 and Google Identity Platform; architected and maintained GCP IAM policy.

- Added automatic fraud detection to the CRM platform, reducing fraudulent activity and improving the bottom line by $75K. Led static application security testing (SAST) integration into CI/CD pipelines.
- Built the GDPR subject access request process, reducing handling time from hours to minutes.

Platform Infrastructure, Terraform & CI/CD

- Enhanced CI/CD by generating Helm charts for Kubernetes deployments, enforcing strict unit testing standards in pull requests, and helping convert hand-built GCP cloud projects to Terraform for repeatable, auditable infrastructure.
- Led migration from self-hosted infrastructure to managed SaaS: Jenkins → Google Cloud Build, RabbitMQ → managed AMQP, self-hosted Elasticsearch → Elastic.co — each planned for correctness, rollback safety, and zero downtime.
- Optimized cloud infrastructure spend by rightsizing workloads, tuning SQL databases to handle increased load with fewer resources ($1K/month saved), and eliminating unnecessary cross-zone network egress costs.

Operational Excellence & Observability

- Designed and rolled out the engineering team's on-call and incident response process from scratch — became the org's "gold standard." Reduced spurious auto-open incidents by 90% through a systematic operational excellence initiative identifying and eliminating the top recurring pain points.
- Championed observability, monitoring, and alerting as first-class requirements for all platform work — not afterthoughts. Established standards for logging, metrics, and deployment hygiene as part of the definition of done.

Distributed Systems & Event-Driven Architecture

- Designed and led delivery of a high-throughput Python 3/Django/Celery email processing platform, scaling from 200K to 1M+ operations/day using RabbitMQ for async worker queuing and Dataflow for batch processing.
- Architected a distributed worker system for high-volume asynchronous workloads, designed for horizontal scalability, fault isolation, and operational visibility.
- Led the rebuild of the core CRM platform (replacing a legacy Java monolith that reached its scaling limit) — scoped the initiative, led architecture reviews, and oversaw delivery across all modules.

AI Tooling & Developer Productivity

- Led team-wide adoption of Anthropic Claude and AI-assisted development tools to accelerate code refactoring, automated test generation, and rapid prototyping. Measurably improved delivery velocity and code quality.
- Mentored engineers at multiple levels; one grew to lead a sibling team. Participated in recruiting, interviewing, and onboarding. Wrote developer documentation and runbooks that de-tribalized platform knowledge and reduced onboarding time.

Squarespace, New York, NY; *Senior Software Development Engineer* April 2019–September 2019

- Contributed to a new microservices-based billing platform. Identified and resolved a critical Kafka serialization defect corrupting billing events in production. Participated in on-call, mentored engineers, and authored system onboarding documentation.

Amazon, New York, NY; *Software Development Engineer* September 2017–February 2019

Helped redesign the distributed architecture for scheduling and configuring content on Amazon Prime Video after the legacy system reached its scaling limit.

- Built Python migration tooling to extract and reformat data from the legacy scheduling system; integrated with DynamoDB for the new architecture.
- Increased Agile rigor and mentored new hires on system design and delivery best practices.

CoreLogic, Knoxville, TN; *Team Lead / Software Development Engineer* April 2015–August 2017

Promoted to team lead for strong backend engineering and production ownership.

- Rebuilt the central code repository server on AWS; scripted full repository migration to GitHub. Resolved critical production defects with regression test coverage.
- Improved application performance by designing a new cache architecture with verified throughput gains. Led security auditing for widely deployed MLS software.

Courion, Westborough, MA; *Staff Software Engineer / Senior Security Software Engineer / Design Lead* December 2005–April 2015

Won 2 President's Club awards (rarely given to engineers) for software delivery and security leadership during the company's migration from on-premises to cloud-hosted SaaS.

- Designed and built identity access management software: permission management, user provisioning/deprovisioning, password reset, and self-service account unlock.
- Led architecture of cryptography and hashing upgrades; owned the full security fix lifecycle from design through test execution and release.

MAPASS, Inc.; *Software Engineer (Founding Engineer)* December 2002–April 2005

- First engineer at an early-stage startup; architected and built the full product from scratch, including third-party integrations, email automation, and a customer-facing web portal.

Mentor Graphics Corporation; *Software Developer* June 2001–December 2002

- Maintained an EDA CAD application; implemented features, fixed bugs, and shipped the installer for a new product release.

EDUCATION

B.S., Computer Science — University of Illinois at Urbana-Champaign, IL (May 2001)